ALIGNMENT MATRIX AUDIT

Curriculum, Assessment, Topic, and ELO Alignment

This report is ETDP's first-pass visual blueprint of how curriculum modules, subjects, topics, assessment criteria, associated assessment-criteria signals, lesson-plan rows, and NQF level descriptors relate inside one qualification.

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QUALIFICATION	CURRICULUM SPINE	TOPIC EVIDENCE FIT	OUTCOME LAYER
91761	**454**	**84%**	**0**
Occupational Certificate: Qualification: Electrician	67 subjects | 454 criteria rows	454 mapped | 0 gap	Extracted ELOs | DB outcomes: 0

Audit Readout

This summary is deliberately honest about what is present, what is inferred, and what is still missing from the formal assessment layer.

- 91761 currently contains 67 subjects, 454 topics, and 454 assessment-criteria rows in ETDP.
- ETDP already holds 302 schedule-ready lesson-plan LPN rows for this qualification, even though the formal LessonPlans table currently contains 454 row(s).
- The outcome layer is incomplete in the database: 0 outcome row(s) and 0 topic-to-outcome links are stored, so the report currently relies on extracted or inferred outcome alignment.
- Summative and formative assessment banks are still missing from ETDP's structured tables: KnowledgeQuestionnaires = 0, Workbooks = 0, LearnerGuides = 0.
- Topic evidence coverage is currently 84%, with 454 mapped topics, 0 developing topics, and 0 gap topics.
- Detected 12 duplicated assessment-criteria cluster group(s), supporting the audit concern that repeated criteria text can mask topic-level curriculum risk.
- The assessment specification file is present and text-readable, so ETDP can now surface that source in the same audit picture instead of relying only on curriculum rows.

Caveats

These cautions protect the report from overstating certainty.

- Exit Level Outcome alignment is currently extracted and heuristically inferred, not yet persisted as a formal database relationship for this qualification.
- No topic rows are explicitly linked to outcome IDs in the database, so subject and module alignment to ELOs should be treated as provisional until the outcome layer is loaded.
- The summative knowledge-question and formative workbook-activity layers are still structurally absent, so this matrix cannot yet prove full assessment-instrument coverage.

Source Document State

The matrix distinguishes files discovered on disk from text that ETDP can actively inspect and reason over.

SOURCE	STATUS	PATH
Curriculum Specification	Available	D:\ETDP\ETDP\Imports\91761\QC_CurriculumSpecification.pdf
Assessment Specification	Available	D:\ETDP\ETDP\Imports\91761\QC_AssessmentSpecification.pdf
Reference Alignment Workbook	Available	D:\ETDP\VocationalLLM\data\knowledge_taxonomy\Alignment_Matrix\alignment_matrix.xlsx
Subject Seed CSV	Loaded	D:\ETDP\ETDP\Imports\91761\CognitiveScan\PipelineJobs\cp-5-20260529115828435\artifacts\normalized_KnowledgeSubjects.csv
Topic Seed CSV	Loaded	D:\ETDP\ETDP\Imports\91761\CognitiveScan\PipelineJobs\cp-5-20260529115828435\artifacts\normalized_KnowledgeTopics.csv

Source materials indexed: 200 | Curriculum text length: 209257 | Assessment text length: 13929

Matrix Layer Status

These counts show which layers of the full alignment stack are already populated.

Subjects	67
Topics	454
Criteria Rows	454
LPN Rows	302
LessonPlans Table	454
DB Outcomes	0
Topic -> Outcome Links	0
Summative Questionnaires	0
Workbooks	0

Extracted Exit Level Outcomes

These are the qualification-level applied competence statements currently extracted from the source documents.

No exit level outcomes were extracted.

NQF Level Descriptor Fit

The qualification is currently marked as NQF level 4.

SCOPE OF KNOWLEDGE

Scope of knowledge, in respect of which a learner is able to demonstrate a fundamental knowledge base of the most important areas of one or more fields or disciplines, in addition to the fundamental areas of study, and a fundamental understanding of the key terms, rules, concepts, established principles and theories in one or more fields or disciplines.

KNOWLEDGE LITERACY

Knowledge literacy, in respect of which a learner is able to demonstrate an understanding that knowledge in one field can be applied to related fields.

METHOD AND PROCEDURE

Method and procedure, in respect of which a learner is able to demonstrate the ability to apply essential methods, procedures and techniques of the field or discipline to a given familiar context, and the ability to motivate a change using relevant evidence.

PROBLEM SOLVING

Problem solving, in respect of which a learner is able to demonstrate the ability to use own knowledge to solve common problems within a familiar context, and the ability to adjust an application of a common solution within relevant parameters to meet the needs of small changes in the problem or operating context with an understanding of the consequences of related actions.

ETHICS AND PROFESSIONAL PRACTICE

Ethics and professional practice, in respect of which a learner is able to demonstrate the ability to adhere to organisational ethics and a code of conduct, and the ability to understand societal values and ethics.

ACCESSING

Accessing, processing and managing information, in respect of which a learner is able to demonstrate a basic ability in gathering relevant information, analysis and evaluation skills, and the ability to apply and carry out actions by interpreting information from text and operational symbols or representations.

PRODUCING AND COMMUNICATING INFORMATION

Producing and communicating information, in respect of which a learner is able to demonstrate the ability to communicate and present information reliably and accurately in written and in oral or signed form.

CONTEXT AND SYSTEMS

Context and systems, in respect of which a learner is able to demonstrate an understanding of the organisation or operating environment as a system within a wider context.

MANAGEMENT OF LEARNING

Management of learning, in respect of which a learner is able to demonstrate the capacity to take responsibility for own learning within a supervised environment, and the capacity to evaluate own performance against given criteria.

ACCOUNTABILITY

Accountability, in respect of which a learner is able to demonstrate the capacity to take decisions about and responsibility for actions, and the capacity to take the initiative to address any shortcomings found.

Module Type Footprint

This mirrors the broad workbook logic: Knowledge, Practical Skill, and Work Experience modules contributing to one qualification picture.

Module modules	1
Module topics	454

Topic Evidence Distribution

Topic evidence is currently the safest quantitative measure because repeated assessment-criteria text can hide real topic gaps.

Mapped 454 | Developing 0 | Gap 0 | Duplicate clusters 12

Topics with evidence	454
Mapped topics	454
Developing topics	0
Gap topics	0

Criteria Duplication Risk

These are the most repeated assessment-criteria clusters currently appearing across multiple topics.

USED IN 18 TOPICS

SE0201 Completed observation checklist | SE0202 Completed section in logbook | SE0203 Completed electrical test report

WA0201 Visually inspect electrical equipment | WA0202 Protection devises used in electrical equipment are inspected and settings verified | WA0203 Test electrical equipment | WA0204 Test main and control circuits | WA0205 Visually inspect installations | WA0206 Electrical tests are done | WA0207 Installation method used is inspected | WA0208 The selection of cables and conductors are inspected

USED IN 15 TOPICS

Electrical installation or control system was correctly and safely isolated according to isolating procedures | Operating procedures and electrical diagrams were correctly interpreted according to manufacturer's speci...

PA0401 Rectify the faults identified | PA0402 Complete a re-commissioning test | PA0403 Report findings and provide feedback | AK0401 Functionality of components | AK0402 Various fault conditions | AK0403 Various control systems | AK0405 Use of testing instruments | AK0406 Interpretation of electrical schematics and drawings

USED IN 14 TOPICS

Describe and explain the application of legislation for working with electrical installations, equipment and appliances | Various types of personal protective equipment are identified and their uses are explained | De...

KT0201 Legislation and regulations for workplace safety within electricity | KT0202 Occupation health and safety legislation | KT0203 Safety precautions and safe practices for working within industry | KT0204 Personal protective equipment | KT0205 Safety symbols and colour coding | KT0206 Fundamentals of isolating and locking out equipment and circuits | KT0207 Fundamentals of securing worksites | KT0208 Protection devices

USED IN 14 TOPICS

Name and describe the types of single phase transformers | Describe, with the aid of drawings, the construction of single phase transformers | Describe and explain the cooling systems used on transformers | Describe,...

KT0301 Theories of single phase wound transformers | KT0302 Types of single phase transformers including single wound, double wound and auto-transformers and their applications. | KT0303 Fundamentals of transformer construction | KT0704 Transformer cooling systems | KT0305 Principles of single phase transformer operation | KT0306 Principles of single phase auto-transformer operation | KT0307 Transformer losses | KT0308 Formulas and calculations on input and output of transformers

USED IN 12 TOPICS

SE0201 Completed checklists | SE0202 Completed section in Logbook

WA0201 Implement safety procedures prior to commissioning | WA0202 Perform functionality testing | WA0203 Monitor instrument readings | WA0204 Rectify defects found | WA0205 Complete a commissioning report | WA0206 Hand over the installation

USED IN 11 TOPICS

Identify different types of electrical measuring instruments including fixed and portable and state the purpose. | Identify the basic components of various measuring and test instruments and describe the basic princip...

KT0201 Portable electrical measuring instruments | KT0202 Types of electrical measuring and testing instruments | KT0203 Safe use of measuring and testing instruments | KT0204 Construction and operating principles of measuring and testing instruments | KT0205 Methods of connecting measuring and testing instruments in circuits | KT0206 Applications and methods of using electrical measuring and testing instruments | KT0207 Care and maintenance of single and three phase measuring and testing instruments | KT0208 Instrument transformers

USED IN 10 TOPICS

Drawings and/or manufacturer specifications were correctly interpreted and the correct installation method was selected that meet the installation specifications, for the given equipment and control systems and compli...

PA0201 Select equipment and control system according to design specifications. | PA0202 Interpret drawings | PA0203 Use manufaturer specifications | AK0201 Relevant subsections of SANS 10142-1 | AK0202 Knowledge of electrical components and switchgear | AK0203 Installation techniques of electrical components and switchgear | AK0204 Read and interpret electrical drawings and installation specifications | AK0205 Knowledge of electronic components

USED IN 10 TOPICS

Electrical installation or control system was correctly and safely isolated according to isolating procedures | Operating procedures and electrical diagrams were correctly interpreted according to manufacturer's speci...

PA0201 Isolate the defective electrical installation or control system | PA0202 Interpret the applicable operating procedure | PA0203 Trace and identify the defect on electrical installation or control system | PA0204 Repair defects on electrical installation or control system | AK0201 Isolating procedures | AK0202 Reading and interpretation of electrical drawings and operating procedure of electrical installation or control system | AK0203 Fault finding procedures and techniques for electrical installation or control system | AK0204 Electrical testing equipment and testing procedures

USED IN 14 TOPICS

Name and describe the types of single phase transformers | Describe, with the aid of drawings, the construction of single phase transformers | Describe and explain the cooling systems used on transformers | Describe,...

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USED IN 10 TOPICS

Risk assessments, including work related hazards and ergonomics, are carried out and reports completed as per statutory requirements. | Evidence of mitigation is observed in accordance with standard operating procedur...

PA0101 Conduct a preliminary risk assessment. | PA0102 Identify the hazards. | PA0103 Relate the risk to the hazard. | PA0104 Conduct mitigation of the risk. | AK0101 Safety procedures | AK0102 Statutory requirements for hazardous conditions/material/substance. | AK0103 Risk assessment procedure for specific context | AK0104 Inspection procedure

USED IN 10 TOPICS

SE0101 Completed observation checklist | SE0102 Completed section in logbook | SE0103 List of equipment installed

WA0101 Apply drawing specifications | WA0102 Use access to height equipment | WA0103 Mark out and do measurements of the installation | WA0104 Drill and/ tap and/or chase walls and/or trenching | WA0105 Install and secure wire ways | WA0106 Mount and fasten equipment | WA0107 Draw in conductors and cables | WA0108 Make off cables

USED IN 10 TOPICS

Work Instructions and testing procedures were correctly interpreted and applied. | Risks were identified and mitigated in accordance with risk management procedures | A test report that confirms the correctness of the...

PA0101 Carry out risk assessment | PA0102 Use the relevant tools and testing equipment | PA0103 Carry out relevant commissioning tests | PA0104 Interpret commissioning tests results | PA0105 Report commissioning results | AK0101 Techniques to carry out commisioning tests | AK0102 Fault finding techniques | AK0103 Isolating and energizing processes

USED IN 10 TOPICS

Work Instructions and testing procedures were correctly interpreted and applied. | Test results were correctly interpreted and recorded in accordance with relevant subsections of SANS 10142-1 and SANS 10242. | A Test...

PA0201 Interpret task instructions | PA0202 Select and use of the correct test instruments | PA0203 Select and use of relevant personal protective equipment | PA0204 Isolate and lock-out of identified equipment. | PA0206 Test electrical equipment, control systems and installations | PA0207 Interpret, record and report test results | AK0201 Relevant subsections of SANS 10142-1 | AK0202 Testing techniques for electrical components and protective devices

Module / Subject / Topic Matrix

Open a module, then a subject, to inspect the actual topic rows, criteria signals, associated assessment-criteria codes, LPN coverage, and topic-evidence score.

Search by module, subject, topic, criteria, IAC code, or outcome...

MODULE

-

Cross-cutting / foundation support | Outcome mapping pending

67	454	302	-
Subjects	Topics	LPN Rows	NQF

KM-01-KT01 - Safety, health, environment, risk and quality principles in the workplace

Indirect support | Cross-cutting / foundation support | confidence 30%

14	14	10	86%	93%
Topics	Criteria	LPN	Evidence	Duplication Risk

KM-02-KT01 - Hand tools and power tools

Unmapped | Outcome mapping pending

3	3	3	84%	0%
Topics	Criteria	LPN	Evidence	Duplication Risk

KM-02-KT02 - Measuring and testing instruments

Unmapped | Outcome mapping pending

11	11	11	90%	91%
Topics	Criteria	LPN	Evidence	Duplication Risk

KM-03-KT01 - Fundamentals of electricity

Indirect support | Cross-cutting / foundation support | confidence 30%

5	5	5	87%	80%
Topics	Criteria	LPN	Evidence	Duplication Risk

KM-03-KT02 - Electronics

Unmapped | Outcome mapping pending

4	4	4	88%	75%
Topics	Criteria	LPN	Evidence	Duplication Risk

KM-03-KT03 - Earthing and bonding

Unmapped | Outcome mapping pending

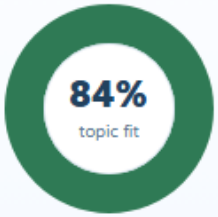
3	3	1	88%	67%
Topics	Criteria	LPN	Evidence	Duplication Risk

KM-04-KT01 - Introduction to the world of work and the electrical trade Indirect support | Cross-cutting / foundation support | confidence 30%	5 Topics	5 Criteria	5 LPN	82% Evidence	0% Duplication Risk
KM-05-KT01 - Wire ways Unmapped | Outcome mapping pending	2 Topics	2 Criteria	2 LPN	84% Evidence	50% Duplication Risk
KM-05-KT02 - Wiring of installations Unmapped | Outcome mapping pending	3 Topics	3 Criteria	3 LPN	82% Evidence	67% Duplication Risk
KM-05-KT03 - Earthing and bonding Unmapped | Outcome mapping pending	3 Topics	3 Criteria	1 LPN	88% Evidence	67% Duplication Risk
KM-06-KT01 - Rotating electrical machinery - AC motors Unmapped | Outcome mapping pending	8 Topics	8 Criteria	2 LPN	82% Evidence	88% Duplication Risk
KM-06-KT02 - Rotating Electrical Machinery - DC motors Unmapped | Outcome mapping pending	7 Topics	7 Criteria	3 LPN	83% Evidence	86% Duplication Risk
KM-06-KT03 - Rotating electrical machinery - Alternators and Generators Unmapped | Outcome mapping pending	7 Topics	7 Criteria	0 LPN	83% Evidence	86% Duplication Risk
KM-07-KT01 - Concepts, theories and principles of supply Systems Unmapped | Outcome mapping pending	10 Topics	10 Criteria	9 LPN	83% Evidence	0% Duplication Risk
KM-07-KT02 - Batteries Indirect support | Cross-cutting / foundation support | confidence 30%	6 Topics	6 Criteria	6 LPN	91% Evidence	83% Duplication Risk

Topic Evidence Coverage

ETDP now measures evidence against topics first, not only against assessment criteria. This keeps the score useful even when the same criteria text is repeated across multiple topics.

Digest Sources + Load Topics | Refresh Topic Evidence | Print / Save PDF | Alignment Matrix Audit



Uploaded sources: 175

Indexed chunks: 4374

Topics with evidence: 454/454

Mapped 454 | Developing 0 | Gap 0

Topic-scale reading

Green means the topic has strong evidence across the uploaded subject matter. Amber means ETDP found partial support but still needs stronger or more varied evidence. Red means Qwen still has a real gap for that topic.

Criteria duplication detected: 12 repeated assessment-criteria cluster(s) were found across topics, so this view uses topic evidence as the main measurement.

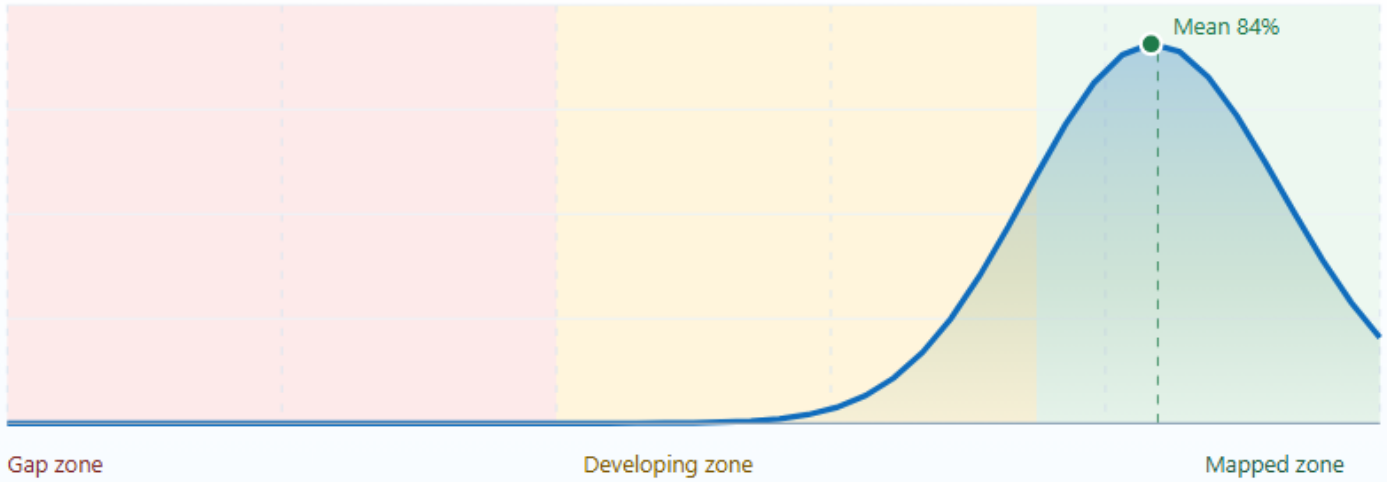
Detected 12 duplicated assessment-criteria cluster(s) across topics, so ETDP is showing topic evidence coverage as the primary measure.

▶ **Show duplicated assessment criteria clusters**

Topic Evidence Bell Curve

Distribution of topic evidence across the qualification

Topic-scale field



Higher density shows where most topic evidence scores are clustering. If the bell leans left, the qualification still has broad evidence gaps. If it shifts right, the uploaded subject matter is carrying more topics into application-ready territory.